

MatSeparate: Evaluating Hierarchical Patch Classifiers for Scene-Level Material Segmentation

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Abstract

Hierarchical material recognition provides a way to classify visual materials at multiple levels of specificity, from coarse physical categories to fine-grained material labels. Recent work by Beveridge et al. introduced the Matador taxonomy and showed strong material recognition from local appearance alone. In this work, we study whether this local hierarchical recognition framework can be extended to material-based segmentation in full scene images. We train local and global-context ResNet50 and hierarchical graph neural network classifiers on Matador-C1, then insert these classifiers into a sliding-window segmentation pipeline evaluated on a curated MINC-S subset. Although our classifiers do not match the Matador-C1 accuracy reported by Beveridge et al., our experiments reveal a broader limitation of local-only material recognition: patches sampled from full scenes often lack object boundaries or contextual cues, making visually similar materials difficult to distinguish. In segmentation experiments, patch-based connected components substantially underperform SAM automatic masks in geometric quality. However, global-context models, especially the global HGNN, improve mapped semantic material accuracy on MINC-S. These results suggest that hierarchical material reasoning is useful for controllable material labeling, but scene-level material segmentation requires global context and stronger spatial grouping mechanisms beyond local patch classification.

1. Introduction

Efficient material-based image segmentation is becoming more important as the capabilities of world-building models widen and demand for higher rendering speeds in complex physical simulations increases. Materially-labeled object masks provide information useful to physical models not only provide more information about an image, but also introduce opportunities to eliminate redundancies in segmentation. For example, depending on the context of a given physical model, different physical resolutions may be appropriate; one model might need to distinguish between a range of textile materials such as cotton, linen, and wool, while another model might only need to label all such objects as "textile." As a separate example, in some physical modeling scenarios it might not be important to distinguish a wooden bowl on a wooden table as two separate objects, so exact object boundaries between the bowl and table would not be necessary to define.

Most existing image segmentation models provide high-resolution object masks but lack material labeling, and those which provide material labels do not provide the flexibility to specify the level of hierarchical detail to which material labeling is necessary [2] [5] [6]. To address this, we leverage the Matador material taxonomy with five hierarchical levels and 57 material labels proposed by Beveridge et al., offering an opportunity to provide more efficient material classification speeds to modeling scenarios which require a lower resolution for material segmentation [3].

In this work, we investigate this gap between local hierarchical material recognition and scene-level ma-

terial segmentation. We begin by attempting to reproduce a strong hierarchical material classifier and study whether global context improves material recognition when local patches are ambiguous. To do this, we train local and global-context flat-ResNet50 and hierarchical graph neural network classifiers on Matador-C1. We then extend this classifier with a sliding-window segmentation pipeline, evaluated on a curated subset of MINC-S, to investigate whether patch-level material predictions are sufficient to produce coherent material masks.

Our results show that hierarchical and global-context models achieve useful material classification across taxonomy levels, but our classifiers do not match the Matador-C1 accuracy reported by Beveridge et al. More importantly, the segmentation experiments reveal that patch-level classification is not enough to recover high-quality material regions. Our connected-component masks substantially underperform Segment Anything model (SAM) automatic proposals in geometric quality [5]. However, global-context models, especially the global HGNN, improve mapped semantic material accuracy on MINC-S. These findings suggest that hierarchical material reasoning remains valuable for controllable material labeling, but accurate scene-level material segmentation requires global context, stronger spatial grouping mechanisms, or end-to-end systems trained on high quality material masks labeled with material hierarchy.

2. Related Work

In 2013, Bell *et al.* released the OpenSurfaces dataset, which provided more than 20,000 real-world scenes labeled with both materials and objects [1]. The same authors released a paper in 2015 providing a framework for material object segmentation using patch classification, along with another dataset called Materials in Context (MINC), which is a collection of images with corresponding materially-labeled object masks [2]. They used a sliding CNN and dense CRF to aggregate pixel-level classes into material object masks, and used the MINC dataset to train, validate, and test their segmentation model. Their best ensemble of GoogLeNet and AlexNet reached 73.1 % mean class accuracy and 79.8 % total accuracy.

Schwartz *et al.* offered a different approach to material image segmentation in 2016 by showing that de-

coupling local views from global views yields better results [6]. They trained one CNN for pixel-level material recognition, then injected context from two separate CNNs, one trained on a local view of the material and the other on the object in its global context. They found that avoid conflating local material information with information about an object’s place in a scene by introducing embeddings from two separate CNNs right before the final prediction layer outperformed a single contextual CNN, and reached per-pixel accuracies of 73.0 % on their own dataset.

While this work was foundational to the field of material image segmentation, it was slightly too early to take advantage of the new and powerful ResNet50, which transformed the computer vision field after its release in late 2015 because of its significantly higher performance on image recognition tasks [4].

An opportunity for sophisticated material segmentation came in 2025 when Beveridge *et al.* introduced a taxonomical material hierarchy, with 57 materials arranged in 5 hierarchical levels [3]. The top-most level is phase (solid/liquid/gas), then the tree moves down through state (biotic/abiotic), composition (metal, rock, etc.), form (e.g. ferrous vs. non-ferrous metal), and finally material. This framework

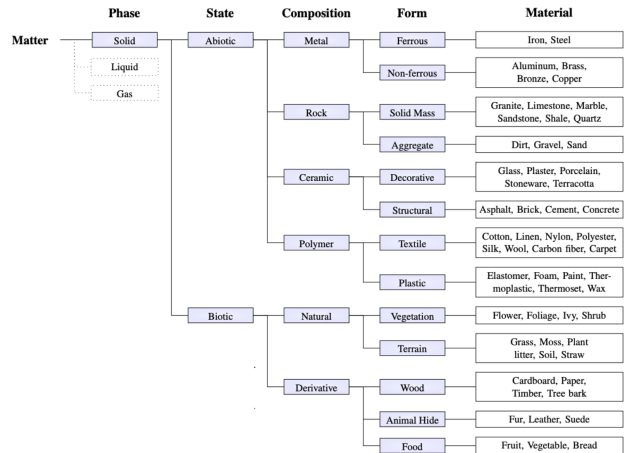


Figure 1: Hierarchical material taxonomy of full Matador dataset. Reprinted from Beveridge et al. (2025)

allows material segmentation classification at higher levels of taxonomy, offering more flexibility in label resolution and opportunities for faster performance.

3. Dataset and Features

3.1. Matador-C1

We trained our patch classifier on a dataset called Matador-C1, which is a subset of the Matador dataset created by Beveridge *et al.* [3]. The original dataset contains 7,200 samples, where each sample contains a local view of a material (e.g. leather, brick, tree bark) along with a corresponding image showing the surrounding global context of the local view. Each sample also contains a black and white view and a LiDAR view of each local view, which we did not use in this project. Following the process of the Beveridge paper, we filtered Matador into Matador-C1, removing materials such as thermoplastic, paint, and glass, that Beveridge *et al.* deemed "insufficient texture to recognize solely from their local appearance" [3]. This left us with 6,614 samples providing coverage over all 37 materials included as leaf nodes in the paper's material taxonomy tree. We use the hierarchical levels in this taxonomy as the material hierarchy for our segmentation model.

3.2. MINC-S

The MINC-S dataset is a segmentation dataset of more than 7,000 material scene segment masks over 1,768 "in-the-wild" scene photos [2]. Each mask is tagged with one of 23 material labels. Since these labels do not align exactly with the 37 material labels used in the Matador dataset, we cropped the MINC-S dataset to 1,067 segment masks over 200 photos, filtering for materials which appear in both datasets (e.g. "pottery" in Matador and "ceramic" in MINC). We use these segment masks to test model performance.

4. Methods

Our system has two stages. First, we train patch-level material classifiers on Matador-C1. Second, we apply these classifiers to full scene images using a sliding-window pipeline that converts patch-level material probabilities into material-labeled connected components. This design lets us test whether hierarchical material recognition from local crops can be extended to scene-level material segmentation.

4.1. Patch-Level Material Classification

We train four patch classifiers: a local flat-ResNet50, a local hierarchical graph neural network (HGNN), a global-context flat-ResNet50, and a global-context HGNN. All models are trained on Matador-C1 and predict labels from the Matador material taxonomy. The local models use only the local material crop, while the global-context models use both the local crop and a corresponding wider patch of the context image.

All image inputs are resized and center-cropped to 224×224 . Images are converted to RGB, scaled to $[0, 1]$, and normalized using ImageNet statistics.

4.1.1 Flat ResNet50 Baselines

Our flat classifier baseline uses a pretrained ResNet50 backbone from the `timm` model library [8]. We replace the final classification layer with a 37-way classifier over Matador-C1 leaf material labels. The model is trained using cross-entropy loss on the leaf class.

The local ResNet50 receives only the local material crop. The global-context ResNet50 uses two separate ResNet50 encoders: one for the local crop and one for the corresponding global context image. The two pooled feature vectors are concatenated and passed through a multilayer classification head:

$$h = [f_{\text{local}}(x_{\text{local}}); f_{\text{context}}(x_{\text{context}})].$$

The final layer predicts the 37 leaf material classes. This model tests whether object-level and scene-level context improve material recognition beyond local texture alone.

4.1.2 Hierarchical GNN Classifiers

The HGNN classifiers are reproduced according to Beveridge *et al.*'s work and predict over all taxonomy nodes rather than only the 37 leaf materials. We represent the Matador taxonomy as a graph whose nodes correspond to material categories at all hierarchy levels, initialized with an average of ResNet50 embeddings of training images belonging to that node (edges represent taxonomy relationships). For each input image, we insert a global graph node with a ResNet50 derived image embedding.

We then apply a two-layer graph attention network

(GAT) to propagate information between the image embedding and the material taxonomy. After message passing, we apply global mean pooling over the graph nodes and use a linear head to produce logits over all taxonomy nodes [7].

The local HGNN initializes the image node from the local crop alone. The global-context HGNN uses separate ResNet50 encoders for the local crop and global context image, concatenates the two feature vectors, projected and inserted into the graph. Figure 2 summarizes the architecture.

4.1.3 HGNN Training Objective

For each training image, the ground-truth label is represented both as a leaf material and as a multi-hot root-to-leaf path through the taxonomy. The HGNN is trained with a combined hierarchical objective. The first term is binary cross-entropy over all taxonomy nodes, encouraging the model to activate every node on the correct path. The second term is a level-wise cross-entropy loss that encourages the model to choose the correct node at each taxonomy depth. Following Beveridge et al., we combine these terms using a greedy maximum. This objective encourages the model to predict the correct fine-grained material when possible while also learning a consistent hierarchy path.

4.2. Patch-Based Material Segmentation

Our segmentation pipeline converts a trained patch classifier into material-labeled connected components. Figure 3 summarizes the pipeline. Given an RGB image, we first sample overlapping square windows using a sliding-window sampler. In the main MINC-S evaluation, we use window size 128 pixels and stride 32 pixels. Each sampled window is resized to 224×224 before classifier inference.

The patch classifier predicts a probability vector over Matador-C1 leaf materials for every sampled window. These predictions are assembled into a coarse material probability grid and bilinearly upsampled to the full image resolution. After interpolation, probabilities are clipped to be nonnegative and renormalized so that each pixel has a valid distribution over leaf classes.

To improve spatial consistency, we refine the dense

probability map using SLIC superpixels. The image is segmented into superpixels, and the material probability vector is averaged within each superpixel. The refined leaf probabilities can then be projected to a chosen taxonomy level. At the leaf level, the projection is unchanged. At an internal taxonomy level, each leaf probability is assigned to its ancestor at that depth, allowing the same probability map to produce segmentations at different hierarchy depths.

Finally, each pixel is assigned to the highest-probability class at the selected taxonomy level. We extract 8-connected components separately for each material class and discard components smaller than 64 pixels. Each remaining component is treated as a predicted material region with a binary mask, material label, bounding box, area, and mean confidence score.

4.3. SAM Baseline

We compare our patch-based material segmentation pipeline against the Segment Anything Model (SAM) as a class-agnostic segmentation baseline [5]. We use the ViT-B checkpoint with automatic mask generation and do not fine-tune SAM. Since SAM does not predict material labels, we evaluate it only as a proposal generator.

For each MINC-S ground-truth material segment, we compute its intersection over union (IoU) with all SAM proposals from the same image and keep the proposal with the highest IoU. This gives SAM oracle credit when any automatically generated proposal overlaps the ground-truth material segment well. Therefore, the SAM comparison measures geometric mask quality rather than semantic material classification.

4.4. Evaluation Protocol

We evaluate segmentation on a curated MINC-S subset containing 200 images and 1067 ground-truth material segments. Each MINC-S annotation is a binary mask with an associated material label. Since Matador-C1 and MINC-S use different material vocabularies, we define a crosswalk between overlapping material classes for semantic evaluation.

For each ground-truth MINC-S segment, we find the predicted component with the highest IoU. This same best-match protocol is used for both our patch-based material segmentation pipeline and SAM auto-

matic proposals. For our pipeline, the matched prediction has both a binary mask and a material label. For SAM, the matched prediction has only a binary mask. As a result, SAM is included in geometric mask-quality metrics but excluded from semantic material-label metrics.

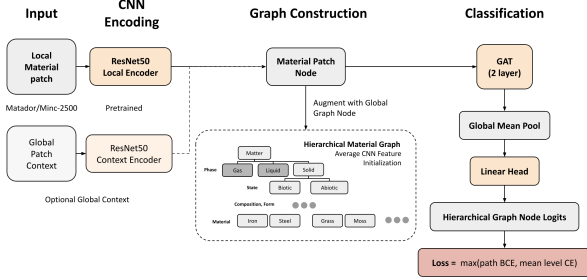


Figure 2: Hierarchy-aware graph classifier. A local or global-context image embedding is inserted as a graph node and aggregated with material taxonomy nodes using graph attention.

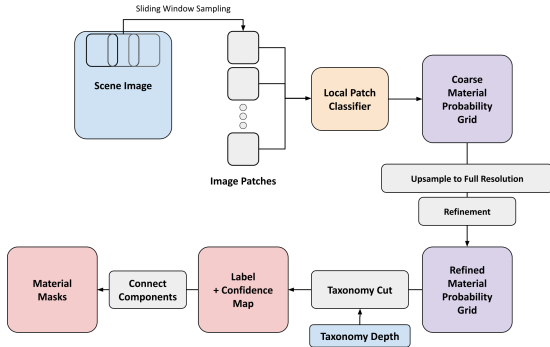


Figure 3: Material segmentation pipelines evaluated in this project.

4.5. Evaluation Metrics

4.5.1 Classifier Metrics

We evaluate patch classifiers using top-1 accuracy at each reported taxonomy level: state, composition, form, and material. Material accuracy is computed directly from the predicted leaf label and the ground-truth Matador-C1 material label. For hierarchy-level

accuracy, predictions are compared after mapping each predicted and ground-truth material label to its ancestor at the corresponding taxonomy level. For HGNN models, internal-level predictions can be read directly from taxonomy-node logits; for flat ResNet50 models, which predict only leaf materials, we map the predicted leaf label to its ancestors.

4.5.2 Segmentation Pipeline Metrics

We evaluate segmentation performance using geometric mask quality, runtime, and semantic material accuracy. For geometric evaluation, each ground-truth MINC-S segment is matched to the predicted component with the highest intersection over union (IoU). We report mean best IoU, which is the average of these best-matching IoU scores across all ground-truth segments.

We also report Recall@0.50, the fraction of ground-truth material segments whose best-matching predicted component achieves an IoU of at least 0.50. In addition, we report the mean number of predicted components per image and the mean inference time in seconds per image.

For semantic evaluation, we report mapped semantic accuracy for the patch-based pipelines. For each ground-truth MINC-S segment, we take the best-overlapping predicted component and check whether its Matador-C1 material label maps to the correct MINC-S material class. SAM is excluded from this metric because its automatic masks are unlabeled.

5. Experiments and Discussion

5.1. Comparing Classifiers

Our first experiment was a comparison between the baseline ResNet50 material patch classification model, and the HGNN-trained material patch classification model that Beveridge suggested, trained only on the local material views in the Matador-C1 dataset. We ran both classifiers on the Matador-C1 dataset, finding that the ResNet50 classifier reached 89.8% leaf accuracy and the HGNN classifier reached 89.1% leaf accuracy. After reaching performance bottlenecks further down the segmentation pipeline, we explored whether incorporating global context into our classifiers would improve performance. We retrained both the ResNet50 and HGNN models on the global views

available in each sample in the Matador database, and found that accuracy dropped to 88.6% leaf accuracy for the ResNet50 classifier but improved to 89.5 % leaf accuracy for the HGNN classifier. Looking solely at leaf-level results, it appears that neither incorporating global context nor adding hierarchical information improves patch classification.

Further up the tree, though, we start to see relative improvements in the global context classifier. Figure 4 shows accuracies at each level of the material hierarchy. These results show that when generalizing to higher levels of the tree, relative performance between the models changes, with global models outperforming local-only models, and the HGNN models outperforming ResNet50. Because the HGNN classifier trains throughout the entire taxonomy, it is better able to predict higher levels of the tree where ResNet, which is only trained on the material labels, fails to do so.

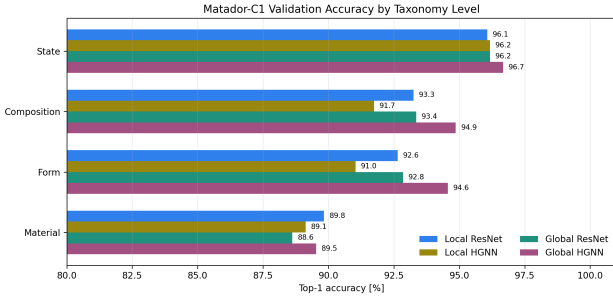


Figure 4: Hierarchical accuracy by taxonomy level for local and global-context classifiers. The models are similar at the leaf material level, while the global HGNN performs strongest at the internal taxonomy levels.

Improvements at higher levels of the tree could be due to the fact that while adding global context is not always enough to provide additional information about the specific leaf-level label for a material, it can yield information which is helpful for broader categorization of a material; for example, the context of being outside in a forest could allow the global classifier to label a low-confidence material object as biotic, while a local classifier would not be able to leverage that information.

Our classifiers show signs of overfitting, as shown

in the training and validation curves for each of them, shown in Figure 5. In particular, training performance continues to improve while validation performance plateaus. This may help explain why our classifiers do not match the Matador-C1 accuracy reported by Beveridge *et al.*

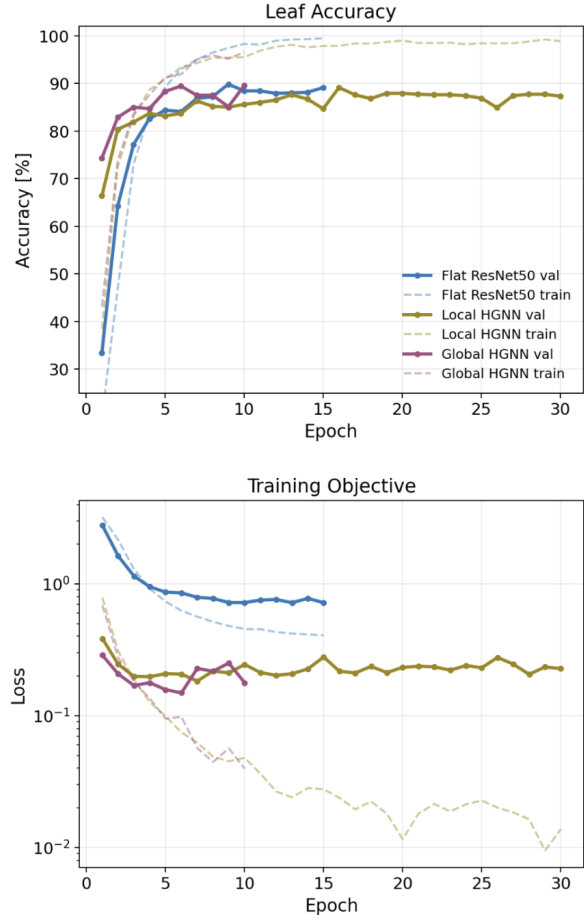


Figure 5: Patch-classifier training curves. Training performance improves while validation performance plateaus, suggesting overfitting on Matador-C1 training examples.

5.2. MatSeparate vs. SAM: Contrasting Segmentation Pipelines

As a baseline, we first run the Segment Anything Model (SAM) on the MINC-S dataset. We then run MatSeparate on MINC-S using all four classifiers trained. The results, shown in Table 1, demonstrate that none of our classifiers are able to reach segmentation performance that is close to competitive with that

of SAM, although our local classifiers run much more quickly.

We can see that based on object masks alone, our

Method	Mean IoU	R@0.50	Comp./Img	Sec/Img
SAM auto	0.661	0.704	63.67	4.459
Local ResNet	0.158	0.035	14.29	2.841
Local HGNN	0.163	0.035	19.62	2.894
Global ResNet	0.191	0.064	22.70	5.133
Global HGNN	0.156	0.037	20.14	5.130

Table 1: Segmentation evaluation on the MINC-S subset of 200 images. SAM automatic proposals remain much stronger in geometric mask quality and the patch-classifier pipelines are competitive only in run-time.

pipeline is suboptimal at zero-shot transfer learning as it lacks the end-to-end training and context necessary to produce high quality image segmentation. However, analysis of the ablations of our segmentation pipelines reveals an interesting finding about the importance of global context in both patch classification and semantic accuracy in segmentation masks.

Mapping semantic accuracy, done by comparing our pipeline’s material label for an object mask to the ground truth MINC-S material label, yields an important result: Global context greatly improves performance in object mask material identification, even though it doesn’t improve performance at the patch classification stage. The global HGNN’s semantic accuracy is greater than three times that of both local models, despite the fact that the local ResNet outperformed the global HGNN in leaf accuracy for patch classification (see Figure 6).

This demonstrates a decoupling between how the tasks of local material classification and semantic patch classification respond to global context. We theorize that this is because imperfections in masks can lend additional global context that global-trained models can take advantage of, whereas local classifiers would only see material overlap within a single mask as noise, and not be able to resolve it since they were trained only on single-material patches.

6. Conclusion

In this paper we presented MatSeparate, a hierarchical material-based segmentation pipeline that converts patch-level material classifiers into material-labeled

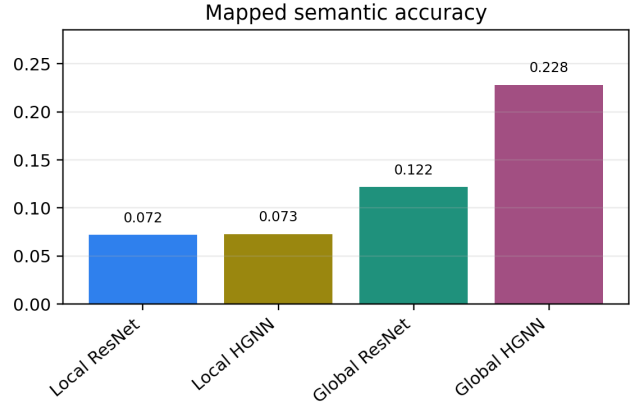


Figure 6: Mapped semantic accuracy of material labels assigned to MatSeparate components on MINC-S. Global-context models transfer better to scene-level material labeling than local-only models.

object masks at a user-specified taxonomy level. Our classifiers achieve strong performance on the Matador-C1 dataset, reaching 96.7% accuracy at the state level and nearly 90% accuracy at the material level. However, we were unable to match the classifier accuracy reported by Beveridge et al., which remains an important limitation of our work.

Although our classifiers performed well, our segmentation pipeline faced serious limitations in its ability to create accurate object masks. Even for our best-performing model, our best mean IoU is only 28.8% that of our baseline. We show that a material-based probability map alone cannot produce well-defined object boundaries, even with the help of methods such as bilinear upsampling and grid refinement. Our model’s only advantage over the baseline was its runtime, with the local classifier-based models running around twice as fast as SAM.

Our main finding is that there is little relation between classification quality and segmentation quality. We theorize that a segmentation pipeline must be trained end-to-end in order to produce high-quality object masks. We attempted to inject a frozen classifier into geometric methods for segmentation, but found that an accurate material-probability map does not on its own produce well-formed object boundaries.

Additionally, we find that the role of global context is much more important at the level of object mask-level semantic identification than it is at the material

classification level. Our results suggest that for scene segmentation global context is not an advantage when classifying local patches, but it is necessary when classifying objects assembled from many local patches, even if all of the local patches are labeled with similar or the same materials.

Our results motivate future work training an end-to-end material segmentation pipeline with a hierarchical patch classifier that is trained directly on material masks. This would lead feedback from unsuccessful boundaries to improve classification strategy, and eliminate the awkwardness of moving between the differing MINC and Matador dataset. Ultimately, our work suggests that progress in material segmentation will not improve with better patch classifiers, and will require architectures that can recognize and segment materials as a single, end-to-end task.

7. Contributions and Acknowledgments

7.1. Group Contributions

Diego Bustamante led the training and eval of the classifiers and co-led writing the report.

Anna Fisher Lopez led the development and implementation of the segmentation pipeline and co-led writing the report.

Ryan Catullo led the eval of the segmentation pipeline and edited the report.

All three members contributed to brainstorming ways to improve performance, creating figures, and analyzing eval results.

7.2. GenAI Usage Statement

We used Claude Opus 4.8 and/or GPT 5.5 to do the following tasks:

- Generate a better README.md file from our codebase

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- Install necessary dependencies in our virtual environment.
- Resolve package inconsistencies and identify the need for additional package imports.
- Write code to load and quality check the Matador and MINC datasets into our repo, and for image and segmentation visualizations.
- Adapt the starter code in Beveridge’s repo for our codebase.